

00_IMPLEMENTATION_PLAN.md

SpeechStudio Implementation Plan

Version: 1.0

Status: Approved

1. Purpose

This document defines the implementation order of SpeechStudio.

The objective is to build the application through small, verifiable iterations.

Each sprint must end with a working application.

No sprint may leave the project in a non functional state.

2. Development Principles

The implementation shall always follow the specification contained in the `spec/` directory.

No feature shall be implemented unless defined by the specification.

Each sprint has a clearly defined scope.

Each sprint must produce a usable application.

Large refactoring between sprints should be avoided.

3. Sprint Workflow

Every sprint follows the same lifecycle.

Read Specification

↓

Implement

↓

Review

↓

Fix Issues

↓

Accept Sprint

↓

Start Next Sprint

No sprint begins until the previous sprint has been accepted.

4. Sprint Overview

Sprint	Goal	Result
Sprint 01	Project Bootstrap	Application starts successfully
Sprint 02	Core Engine	Backend operational
Sprint 03	GUI Framework	Complete application shell
Sprint 04	Voice System	Voice cloning workflow
Sprint 05	Prompt System	Full Higgs prompt control
Sprint 06	Speech Generation	End to end speech generation
Sprint 07	Polish	Production ready application

5. Sprint 01

Name

Project Bootstrap

Goal

Create the complete runtime environment.

Scope

Project structure

Virtual environment

Dependency installation

Folder creation

Hugging Face cache

Model download

Tokenizer download

Application launcher

Acceptance Criteria

Application starts.

Environment is created automatically.

Model download works.

No manual Python commands required.

Definition of Done

Double clicking launch.bat starts SpeechStudio.

6. Sprint 02

Name

Core Engine

Goal

Implement the backend foundation.

Scope

ModelManager

PromptBuilder

VoiceManager

GenerationManager

AudioManager

SettingsManager

HistoryManager

Logger

Acceptance Criteria

Engine starts.

Model loads.

Tokenizer loads.

Engine exposes internal API.

Definition of Done

Backend is operational.

No speech generation yet.

7. Sprint 03

Name

GUI Framework

Goal

Create the complete application shell.

Scope

Main Window

Menu

Toolbar

Sidebar

Prompt Editor

Control Panel

Status Bar

Waveform placeholder

Acceptance Criteria

Every planned panel exists.

Layout is resizable.

Dark theme active.

Definition of Done

Complete GUI structure exists.

Backend connection ready.

8. Sprint 04

Name

Voice System

Goal

Implement voice profile management.

Scope

Voice Library

Voice Import

Voice Export

Reference Audio

Reference Transcript

Voice Validation

Acceptance Criteria

Voice profiles can be created.

Voice profiles can be selected.

Reference audio loads correctly.

Definition of Done

Voice cloning workflow prepared.

9. Sprint 05

Name

Prompt System

Goal

Implement the complete Higgs prompt builder.

Scope

Emotion

Style

Prosody

Sound Effects

Prompt Preview

Prompt Validation

Acceptance Criteria

Every official Higgs token available.

Prompt Preview shows final prompt.

Validation detects invalid combinations.

Definition of Done

Prompt generation fully operational.

10. Sprint 06

Name

Speech Generation

Goal

Generate speech from the GUI.

Scope

Generate button

Stop button

Generation parameters

Audio playback

History

Output management

Acceptance Criteria

User enters text.

Selects voice.

Selects emotion.

Presses Generate.

Speech is produced.

Definition of Done

Complete speech generation workflow operational.

11. Sprint 07

Name

Polish

Goal

Finalize the application.

Scope

Settings

History improvements

Benchmark

Logging

Performance

Error dialogs

UI refinements

Acceptance Criteria

Application behaves consistently.

No major known issues.

Portable operation verified.

Definition of Done

Version 1.0 ready.

12. Sprint Review

After every sprint the following questions shall be answered.

Does the application still start?

Does every previously completed feature still work?

Does the implementation match the specification?

Were unnecessary dependencies introduced?

Was portable operation preserved?

13. Change Policy

During implementation the specification remains stable.

If a required change is discovered

Stop implementation.

Update the specification.

Review the change.

Resume implementation.

Implementation shall never silently diverge from the specification.

14. Definition of Project Completion

SpeechStudio Version 1.0 is complete when

All seven sprints are accepted.

All specification documents are implemented.

The application is fully portable.

The application runs locally.

The application exposes every supported Higgs Audio V3 capability through the graphical interface.

No Python editing is required by the end user.